



Proyect name: “CallMeMaybe”

Analyst: Michelle Tirado

January 1st, 2026

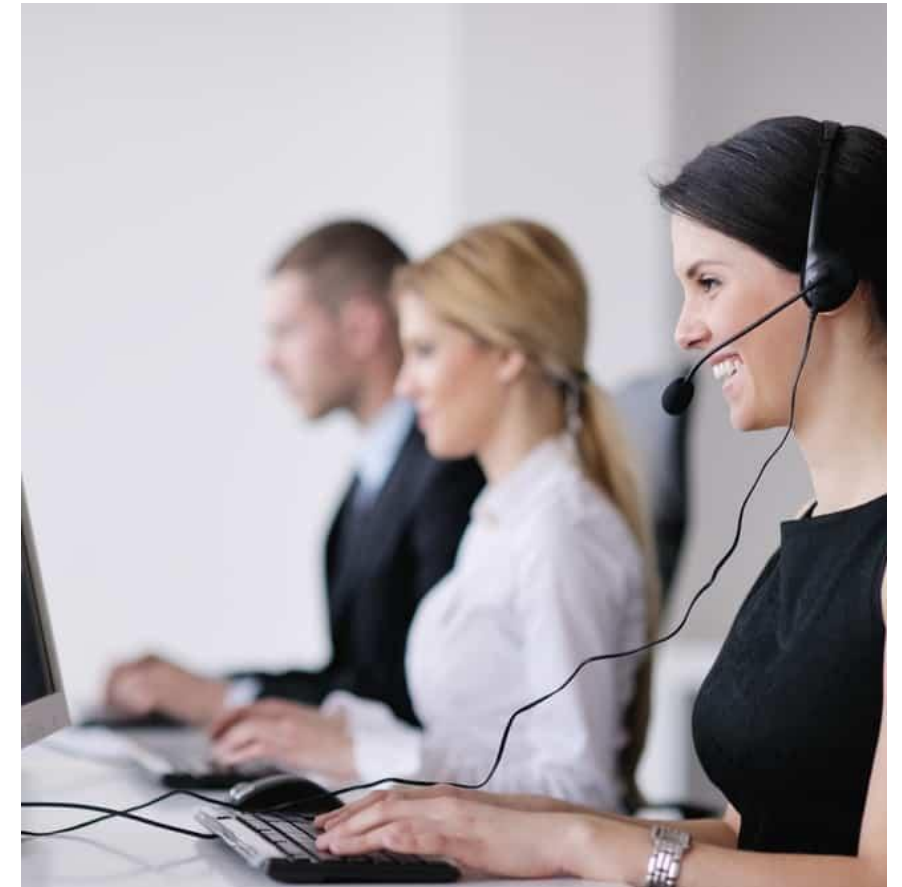
Project Information

Date: January 2nd, 2026.

Analyst name: Michelle Tirado

Project name: “CallMeMaybe”

Project objective: Analyze agent performance metrics to identify top-performing and underperforming agents.

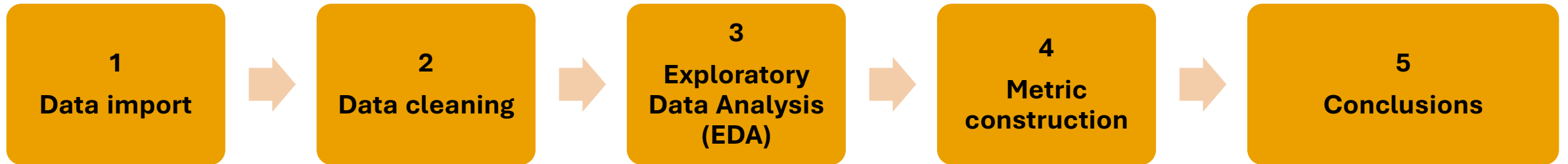


RESUME

This business report presents an analysis of agent performance in a call center environment. The tables, figures, analyses, and conclusions are derived from the data analysis. The data were imported from a CSV file and cleaned to remove duplicate records and empty cells. The cleaned dataset was then analyzed to examine client behavior, operator performance, and to calculate key performance indicators (KPI).



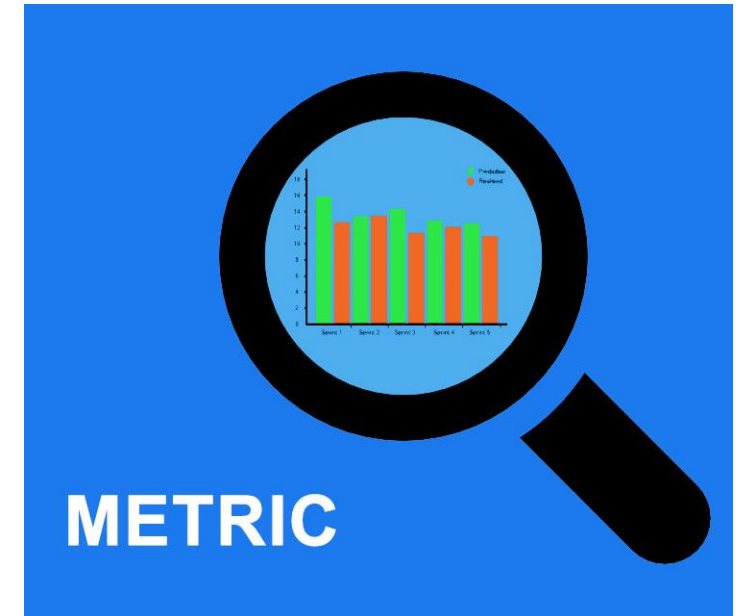
PIPELINE FOR DATA ANALYSIS



DATA DESCRIPTION

The “telecom_dataset_us.csv” dataset contains information about the usage of the “Call Me Maybe” virtual telephony service. The customers are organizations that need to distribute large volumes of inbound calls among multiple operators. Operators can also place internal calls to communicate with one another. All calls are routed through the **Call Me Maybe** network. The compressed dataset telecom_dataset_us.csv includes the following columns:

- **user_id**: Unique identifier of the customer (client) account.
- **date**: Date on which the statistics were recorded.
- **direction**: Call direction, either inbound (in) or outbound (out).
- **internal**: Indicates whether the call was internal (between operators) or external (with a customer).
- **operator_id**: Unique identifier of the operator handling the call.
- **is_missed_call**: Indicates whether the call was missed (True) or successfully answered (False).
- **calls_count**: Number of calls recorded on the given date.
- **call_duration**: Duration of the call in seconds, excluding waiting time.
- **total_call_duration**: Total call duration in seconds, including waiting time.



1. DATA IMPORT

The dataset contains 8 columns and a total of 53,902 entries. The "operator_id" and "internal" columns contain missing values, which require further investigation during the data-cleaning process. In addition, some columns have incorrect data types that need to be corrected as part of data cleaning.

```
Columns with empty values
Out[3]:
→ operator_id      8172
→ internal         117
user_id            0
direction          0
date               0
is_missed_call     0
calls_count        0
call_duration      0
total_call_duration 0
```

2. DATA CLEANING

This phase includes: A) Cell format correction, B) Removing empty cells, and C) Eliminating duplicates.

The original data has:

Total entries: 53,902

Total users: 307

Total operators: 1,092

Total number of recorded dates: 119

Missing information:

Total empty entries: 8,172

Total number of affected users: 305

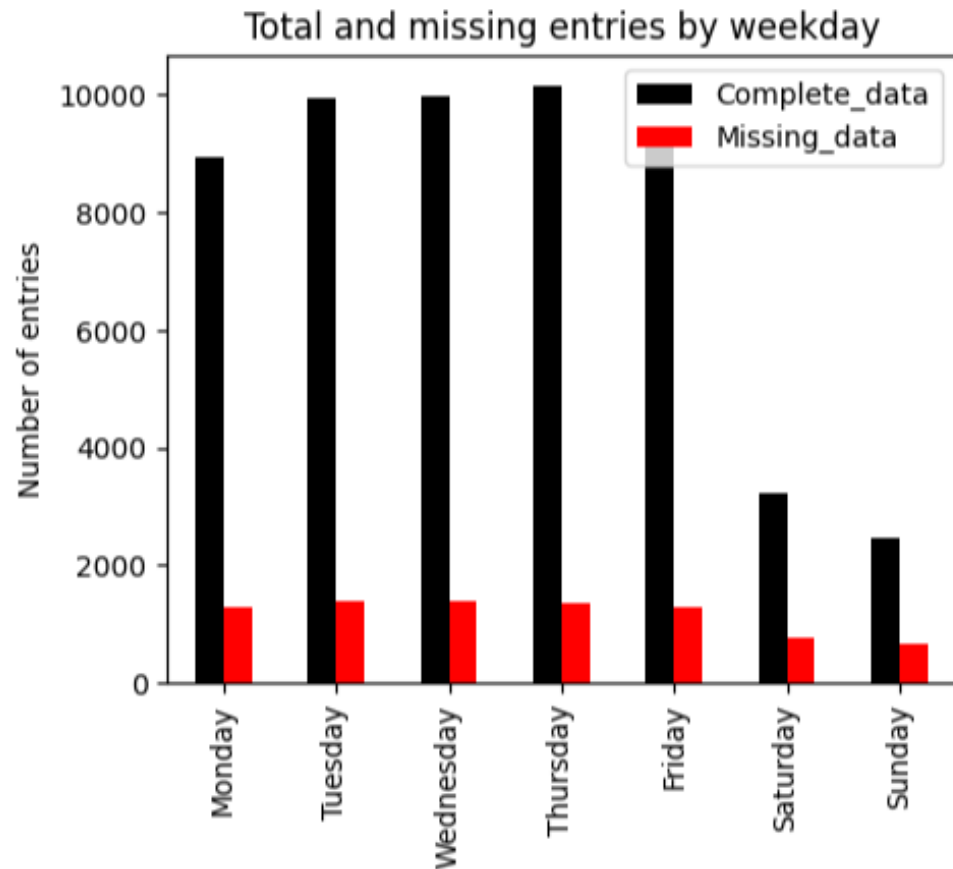
Total number of affected days: 119

Duplicates:

Duplicated cells: 4,179

Empty cells were found in the **operator_id** field, which is a categorical variable. Since these values cannot be replaced, the corresponding records were removed.

The missing entries represent 15.16% of the total entries. To identify the most impacted days, a graph was constructed. Call volume is higher on weekdays compared to weekends; consequently, the amount of missing data is greater on these days. However, when calculating the proportion of missing cells recorded per day (named missing rate), it becomes evident that **weekends have the highest missing data rates.**



	Missing rate (%)
Monday	14.26
Tuesday	14.13
Wednesday	14.03
Thursday	13.48
Friday	14.01
Saturday	23.53
Sunday	27.40

CLEAN DATASET

The cleaned dataset was reduced to **41,491** entries.

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 41491 entries, 1 to 53899
```

```
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
----	-----	-----	-----
0	user_id	41491 non-null	object
1	date	41491 non-null	datetime64[ns, UTC+03:00]
2	direction	41491 non-null	object
3	internal	41491 non-null	object
4	operator_id	41491 non-null	Int64
5	is_missed_call	41491 non-null	bool
6	calls_count	41491 non-null	int64
7	call_duration	41491 non-null	int64
8	total call duration	41491 non-null	int64

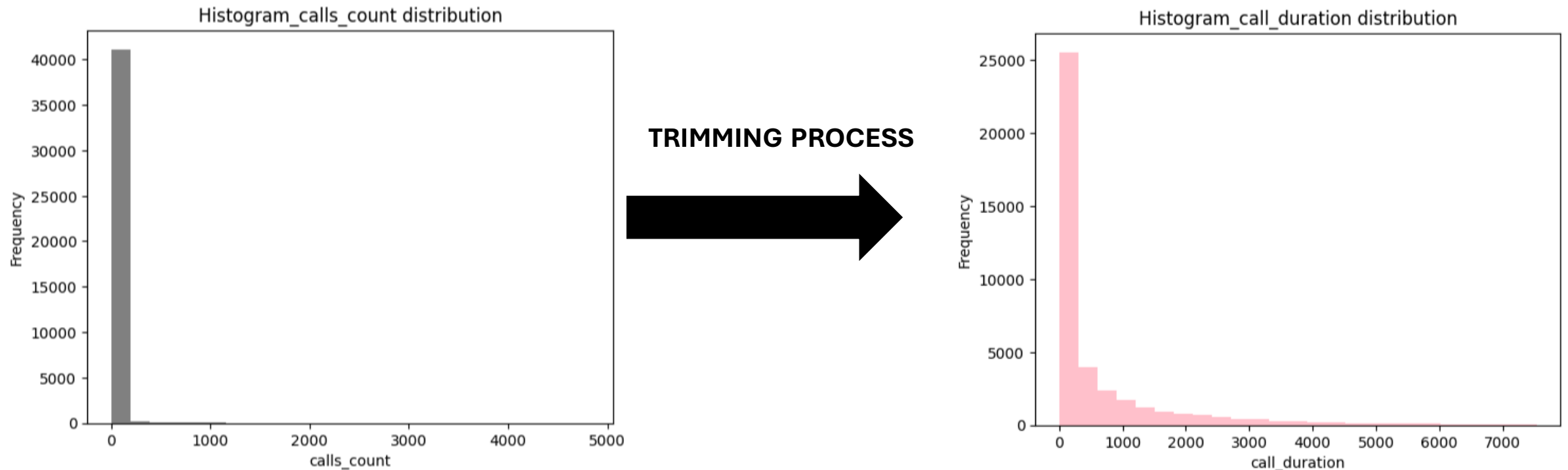
3. EXPLORATORY DATA ANALYSIS (EDA)

Basic statistic was used to explore data, and the greatest finding are presented:

- The minimum date recorded in the dataset is 2019-08-02
- The maximum date recorded in the dataset is 2019-11-28
- The date range recorded in the dataset is 118 days
- The minimum calls count recorded by agent is 1
- The maximum calls count recorded by agent is 4,817
- The average calls count is 17
- The median calls count is 4.0
- The minimum call duration recorded in the dataset is 0 seconds
- The maximum call duration recorded in the dataset is 144,395 seconds (40 hours)
- The call duration mean recorded in the dataset is 1,011 seconds (17 minutes)
- The call duration median recorded in the dataset is 106 (~ 2 minutes)
- The minimum waiting duration recorded in the dataset is 0 seconds
- The maximum waiting duration recorded in the dataset is 46,474 (13 hours)
- The waiting_duration mean recorded in the dataset is 312 seconds
- The waiting_duration median recorded in the dataset is 60 seconds



The exploratory analysis reveals a **highly right-skewed distribution**. For example, the maximum waiting duration recorded in the dataset is 46,474 seconds (approximately 13 hours), which makes no sense, and likely indicates a failure in the recording system for that observation. Therefore, a data **trimming process** was applied to remove extreme values in *calls_count*, *call_duration*, and *total_call_duration* as all these variable were affected. As a result, only the 0–99th percentile of the data was retained to reduce the impact of highly right-skewed distributions.

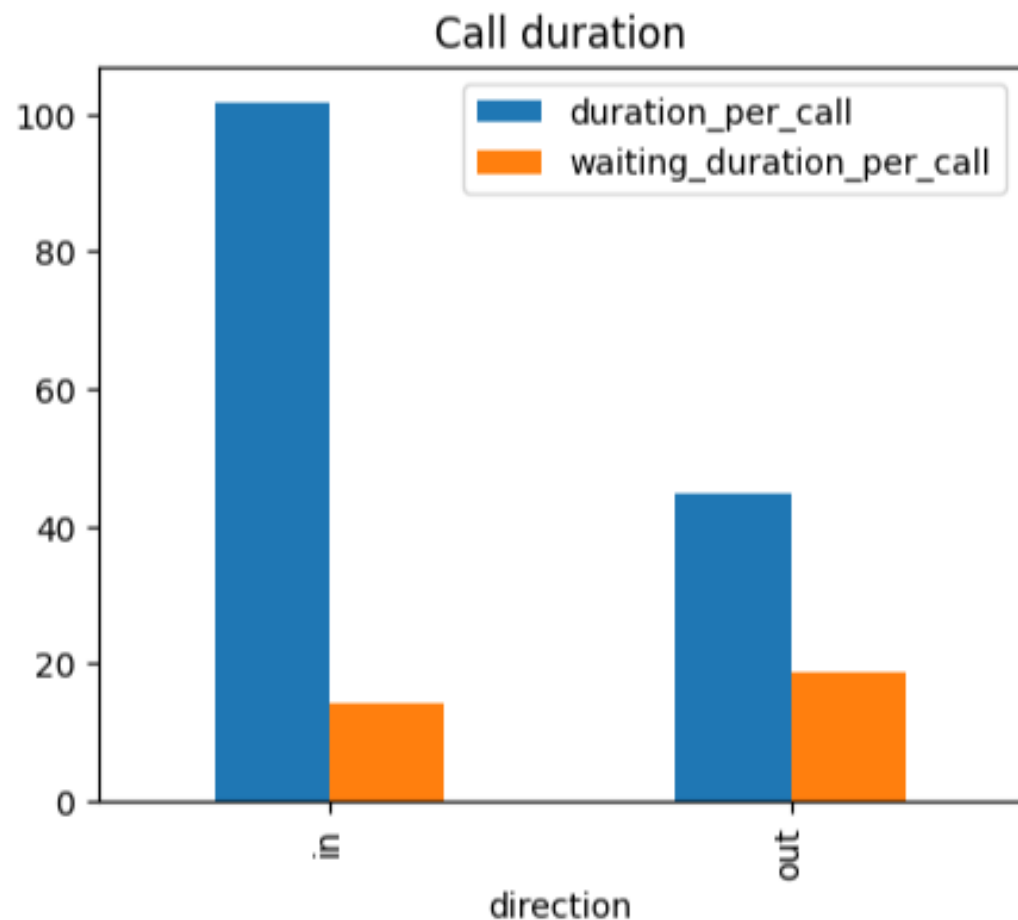


RESULTS OF THE TRIMMING PROCESS

INSIGHTS

DATASET		TRIMMED DATASET
The maximum number of calls recorded per agent per day in the non-trimmed dataset is 4,817.		The maximum number of calls recorded per agent per day in the trimmed dataset is 164.
The longest call recorded in the non-trimmed dataset is 40.11 hours.		The longest call recorded in the trimmed dataset is 2.09 hours.
The longest waiting time recorded in the non-trimmed dataset is 12.91 hours.		The longest waiting time recorded in the trimmed dataset is 1.56 hours.

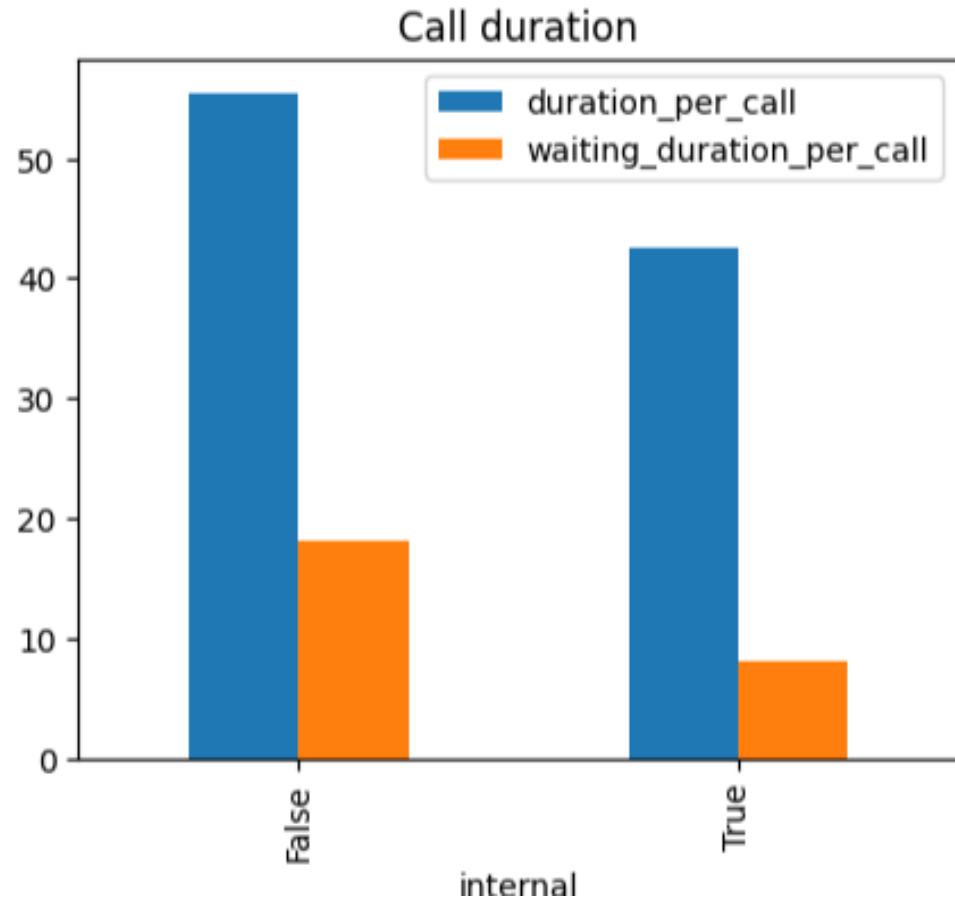
IINBOUND CALLS VS OUTBOUND CALLS



	INBOUND	OUTBOUND
Total calls	80,535	367,885
Duration per call (seconds)	102	45
Waiting time per call (seconds)	14	19

In the database, there is a record of 17.96% incoming calls and 82.04% outbound calls.

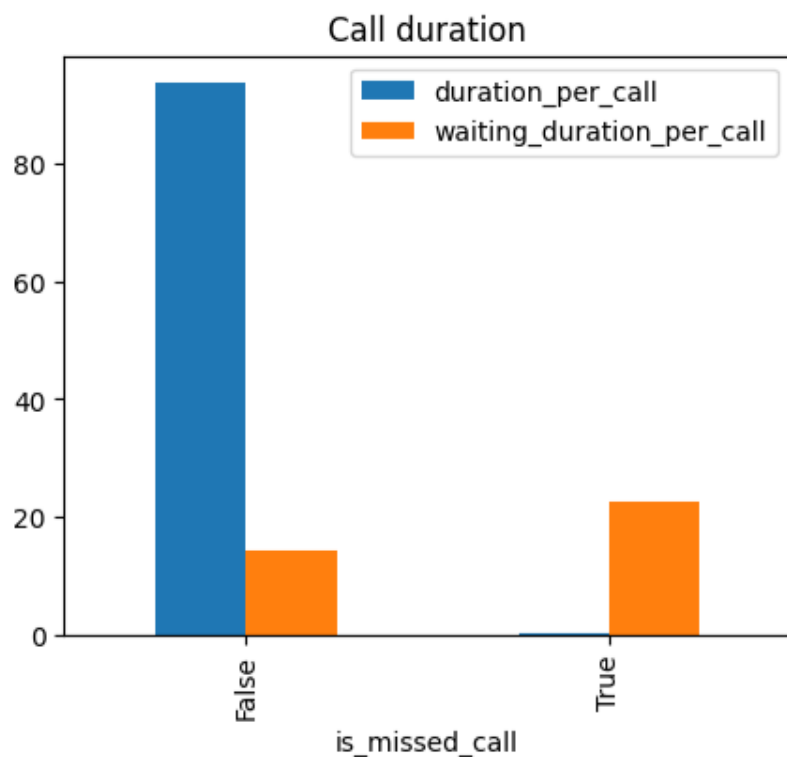
INTERNAL VS EXTERNAL CALLS



	EXTERNAL	INTERNAL
Total calls	434,755	13,665
Duration per call (seconds)	56	18
Waiting time per call (seconds)	43	8

There is 96.95% external calls and 3.05% internal calls

MISSED CALL VS ANSWERED CALL



	MISSED CALL	ANSWERED CALL
Total calls	184,962	263,458
Duration per call (seconds)	0	94
Waiting time per call (seconds)	23	14

There is 58.75% received calls and 41.25% missed calls.

Exploratory Data Analysis Insights

- Overall, operators spend more time on outbound calls than on inbound calls. The total call duration for outbound calls is approximately twice that of inbound calls.
- Most calls (96.95%) involve external sources (clients). Of all calls, 58.75% are answered and 41.25% are missed. Missed calls have a higher average waiting time (23 seconds) compared to answered calls (14.36 seconds).
- On average, an operator spends 93.69 seconds per call, with an average waiting time of 14.36 seconds. Only 63.54% of calls last longer than 10 seconds.

4. METRIC CONSTRUCTION

Using the trimmed dataset, **Key Performance Indicators (KPIs)** will be constructed. Based on standard teleperformance KPIs found in external bibliographic resources, a dataset containing these indicators will be built. The selected metrics are:

- **Total calls**
- **Average Handle Time (AHT):** Average time per call (seconds)
- **Average Waiting Time (AWT):** Average waiting time per call (seconds)
- **Abandon Rate (AR):** Is calculated as `missed_calls` divided by `total_calls`, with values ranging from 0 to 1.

Agent performance is evaluated using AHT, AWT, and AR, where lower values indicate better performance, while higher values reflect poorer performance. Because this metrics are independent variables, they will be standardized to be summed and create a **performance index**. The proposed formula is:

$$\text{Performance Index} = zAHT + zAWT, + zAR$$

Where:

$zAHT$ = standarized Average Handle Time

$zAWT$ = standarized Average Handling Time

zAR = standarized Abandon Rate

To make this report more effective, only the 10 highest- and lowest-performing operators are presented. The calculated metrics for all operators are included in the “agents_performance.xlsx” file.



10 BEST PERFORMANCE OPERATORS

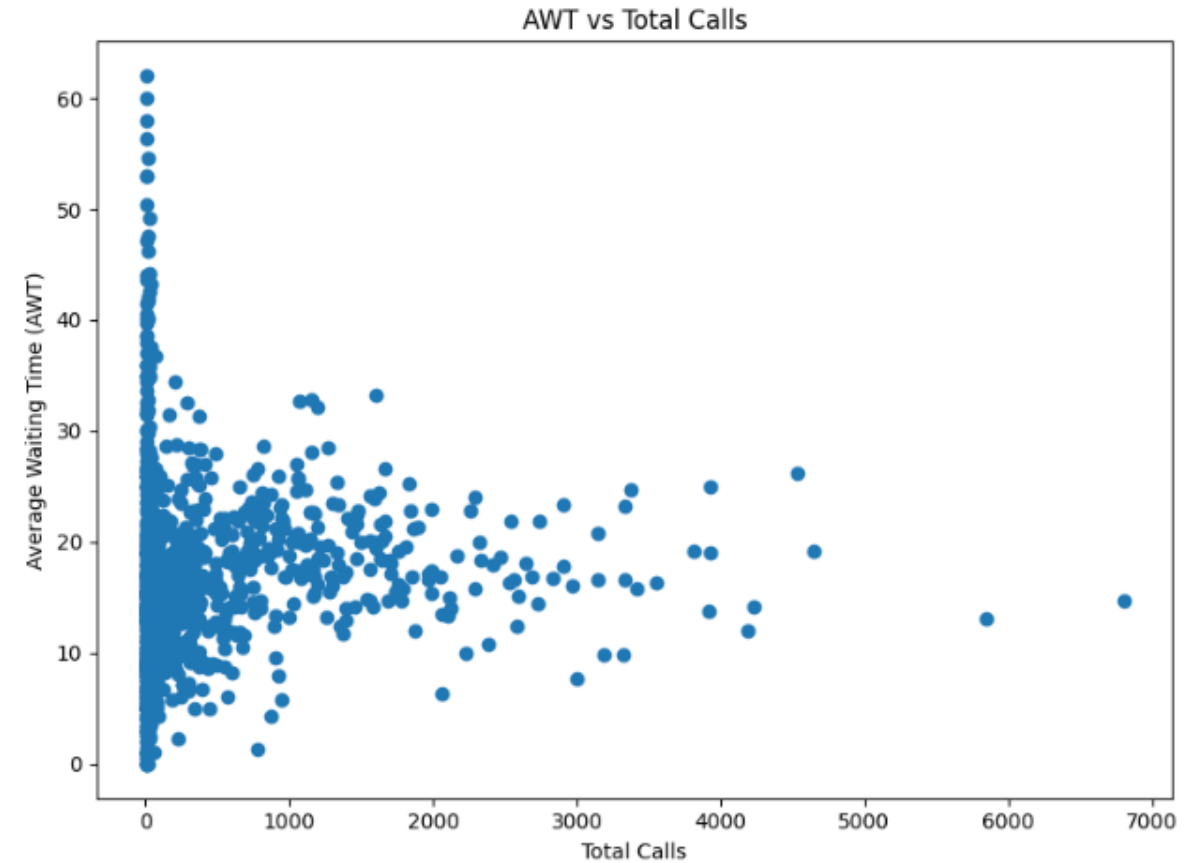
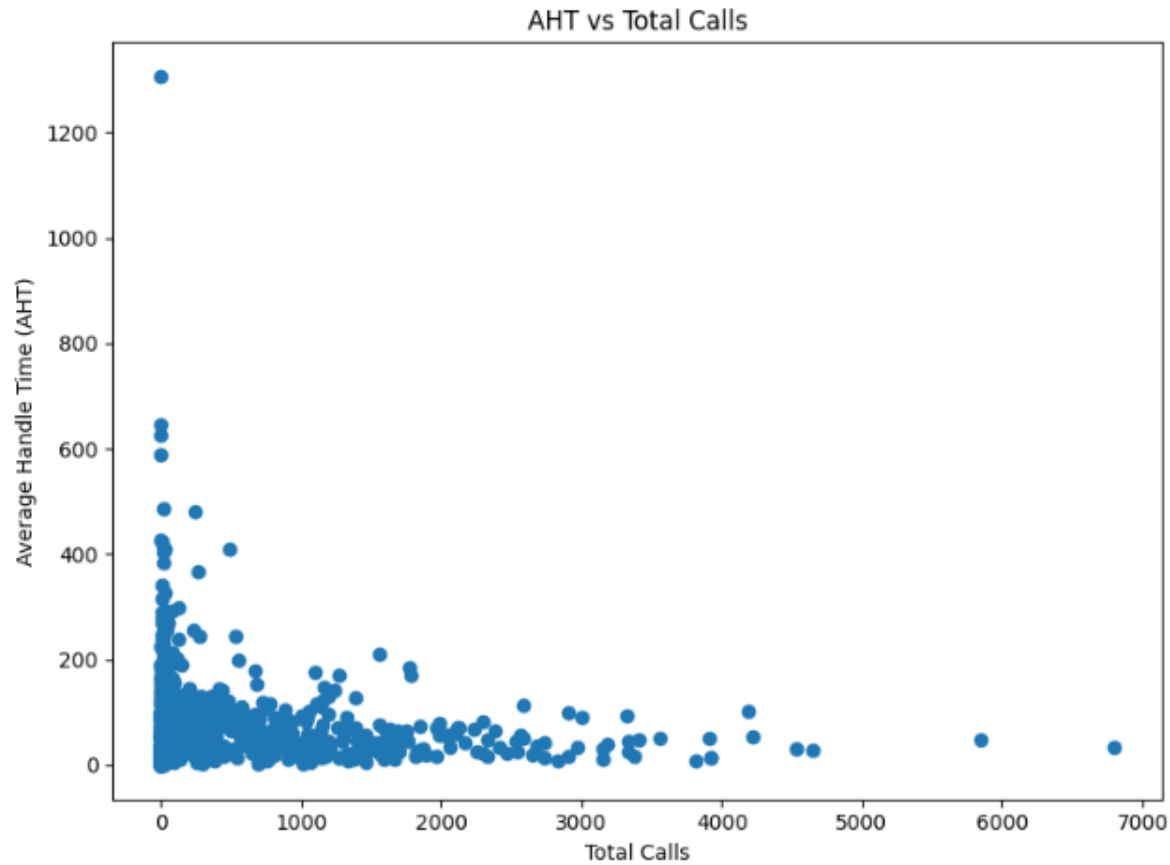
operator_id	total_calls	AHT	AWT	AR	performance_index
968520	5	6.60	1.00	0.0	3.85
952954	2	7.50	1.00	0.0	3.84
930692	6	9.67	0.83	0.0	3.84
954086	32	9.91	2.25	0.0	3.66
946020	1	5.00	3.00	0.0	3.63
896020	16	29.50	0.94	0.0	3.57
952968	3	10.00	4.33	0.0	3.41
952982	8	48.88	0.88	0.0	3.33
891948	1	33.00	3.00	0.0	3.28
884294	1	5.00	6.00	0.0	3.27

10 LOWEST PERFORMANCE OPERATORS

operator_id	total_calls	AHT	AWT	AR	performance_index
891192	1	1306.00	32.00	0.00	-16.42
932246	1	0.00	60.00	1.00	-6.88
899898	2	590.00	25.00	0.00	-6.47
899906	1	626.00	19.00	0.00	-6.20
917890	1	647.00	13.00	0.00	-5.74
899900	2	191.00	58.00	0.00	-5.37
956080	5	237.40	50.40	0.00	-5.05
909768	14	278.64	42.00	0.00	-4.56
891154	487	408.52	18.16	0.32	-4.52
918390	246	479.75	15.65	0.13	-4.42

As shown in the AHT graphic, users' calls tend to last between 0 and 200 seconds.

As shown in the AWT graphic, users tend to wait between 0 and 30 seconds before their calls are answered.



CONCLUSIONS

The original dataset contained 53,902 entries. A total of 8,172 rows with missing values in the "operator_id" and "internal" columns were removed, and an additional 4,179 rows were removed due to duplicated records. The missing entries were analyzed to determine whether the missing data was associated with a specific date or client; however, no specific pattern was identified. Weekends exhibited the highest missing-value rates. Overall, the missing entries represented 15.16% of the total dataset and were removed from the analysis. After the data-cleaning process, the dataset was reduced by 23.03%, resulting in a final dataset of 41,491 entries.

The dataset presents the records obtained from 2019-08-02 to 2019-11-28 (118 days).

The clean dataset was trimmed as inconsistent extreme values were detected.

The operators spend more time on outbound calls than inbound calls.

There is 58.75% received calls and 41.25% missed calls. The missed calls has an average record of 23 seconds on waiting time, which is higher than the 14.36 seconds recorded for non-missed calls.

Only 63.54% of the calls last more than 10 seconds, contributing to the observed call duration distribution (right-skewed).

REFERENCIAS USADAS PARA EL ANÁLISIS

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